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Stuck Pipe Detection in Oil and Gas Drilling Operations Using Deep Learning Autoencoder for Anomaly Diagnosis

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Abstract: Stuck pipe events remain a critical challenge in oil and gas drilling operations, leading to increased non-productive time and substantial financial losses. Traditional detection methods rely on manual monitoring and expert judgment, which are prone to delays and human error. This study proposes a deep learning autoencoder-based anomaly diagnosis approach to enhance the detection of stuck pipe incidents. Using high-resolution time series drilling data from the Volve field, a deep learning autoencoder model was trained exclusively on normal drilling conditions to learn operational patterns and detect deviations indicative of stuck pipe events. The proposed model leverages reconstruction error as an anomaly detection metric, effectively distinguishing between normal and stuck cases. The results demonstrate that the model achieves a detection accuracy of 99.06%, with an area under the receiver operating characteristic curve (AUC) of 0.958. Additionally, the model attained a precision of 97.12%, a recall of 91.34%, and a F1-score of 94.15%, significantly reducing false positives and false negatives. The findings highlight the potential of deep learning-based approaches in improving real-time anomaly detection, offering a scalable and cost-effective solution for mitigating drilling disruptions. This research contributes to advancing intelligent monitoring systems in the oil and gas industry, reducing operational risks, and enhancing drilling efficiency.

Keywords: stuck pipe; drilling; LSTM; deep learning

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1. Introduction

1.1. General Background

Drilling operations in the oil and gas industry are highly complex and require precise control to ensure safety, efficiency, and cost-effectiveness [1–7]. One of the most important problems encountered in drilling is pipe sticking, where the drill string becomes lodged in the wellbore and becomes stuck and unable to proceed any further [8–12]. Pipe sticking can lead to significant delays in operations, additional costs, and abandonment of the well in extreme situations if left unmanaged. Pipe sticking incidents are caused by a number of factors, including geological conditions, wellbore instability, differential pressure, and improper drilling practices. With the advancement in complexity and depths in modern day drilling activities, particularly in offshore and unconventional reservoirs, avoiding the hazards of pipe sticking have become a priority for the industry [13–18]. Pipe sticking carries implications beyond financial loss, as it affects the efficiency and safety of the entire drilling process. In the event of pipe sticking, operators are forced to undertake costly remedial measures such as jarring, backreaming, or side-tracking the wellbore in an attempt to free the stuck pipe. These remedial measures lead to delays in operations,

as well as create risks of formation damage, failure of tools, and additional wear on the drill components. Pipe sticking also leads to non-productive time (NPT), an important parameter in measuring the efficiency of a drill process. Reducing NPT becomes essential in order to maintain cost efficiency, as the costs of drilling continue to rise with increasing depths and challenging well conditions.

Several factors contribute to pipe sticking, with mechanical and differential sticking being the most common types. Mechanical sticking results from the physical blockage of the drill string due to wellbore instability, cuttings accumulation, or key seat formation [19–22]. Differential sticking, however, results from pressure differentials between the drill pipe and the surrounding formation, causing the pipe to adhere to the wellbore wall. Inefficient hole cleaning, high mud weight, and poor drilling fluid properties are some of the factors that can exacerbate the risk of pipe sticking. The interaction of these factors needs to be understood for the formulation of effective prevention and mitigation strategies. With growing technical difficulties in drilling operations, traditional stuck pipe prevention and detection techniques have been found lacking in many cases. Traditional techniques rely on real-time surface monitoring, historical data analysis, and drilling engineer experience to identify early warning signs. While these techniques have been useful, they are often beset by late detection, subjective interpretations, and weak predictive abilities. The dynamic drilling process requires more proactive and data-driven approaches to enhance early detection, improve decision-making, and reduce operational risks associated with stuck pipe incidents.

Recent advancements in stuck pipe detection have extended beyond traditional mechanical indicators that incorporated alternative technologies of physical modeling, sensor-based diagnostics, and hybrid signal analysis. For example, magnetic flux leakage analysis has been proposed to detect outer wall defects in bimetal composite pipes with high sensitivity [23]. In a related study, contact property evaluation and friction modeling were used to enhance the predictive understanding of pipe string behavior during pipe jacking operations [24]. Additionally, non-invasive diagnostic techniques, such as radioisotope-based blockage localization in multiproduct pipelines, have been successfully employed to detect obstructions with precision [25]. These alternatives reflect the diverse technological landscape in which stuck pipe detection strategies continue to evolve and strengthen the need for new methodologies.

In parallel, explainable artificial intelligence (xAI) has gained prominence for enhancing trust and interpretability in machine learning applications. xAI frameworks provide predictions and clarify the reasoning behind them, which increases transparency. While primarily explored in cybersecurity and industrial threat detection [26,27], xAI has recently been integrated into physical diagnostics as well, such as battery health estimation using ultrasonic signals [28]. These developments highlight the potential for combining explainable AI with deep learning-based anomaly detection models in drilling, which paves the way for future recommendations.

1.2. Challenges in Traditional Detection Methods

The detection and prevention of stuck pipe incidents in oil and gas drilling have historically relied on conventional monitoring techniques that depend on real-time surface data, historical well records, and the expertise of drilling engineers [29–31]. These methods rely on the measurement of variations in weight on bit, torque, hook load, rate of penetration, and standpipe pressure to sense anomalies representative of pipe sticking. While such indicators can be valuable in providing insight, they are typically reactive and non-predictive in nature and, consequently, only identify occurrences of a sticking pipe once they have occurred or when mitigation becomes increasingly difficult. Manual obser-

vation also involves subjectivity and variability, as different engineers may have different interpretations of the same information and, consequently, response times may be delayed. Apart from this, conventional methods of sticking pipe detection are incapable of coping with the non-linearity and dynamics of the interactions between the drilling parameters, the formation characteristics, and the fluid dynamics, and consequently, early diagnosis becomes very challenging. Due to the fact that the conditions of drilling are very different in different well environments, surface-based observation by itself typically fails to capture downhole complexities, leading to sticking pipe occurrences.

Traditional detection methods lack real-time predictive capability, and they lead to longer NPT and higher cost. Due to the reactive nature of these methods, they are inappropriate for demanding drilling, e.g., in directional and extended-reach wells. Historical data analysis is undermined by availability and quality and might not reflect the current conditions. Deeper formations also complicate detection with pressure, temperature, and geomechanical uncertainties. Furthermore, the vast, multidimensional drilling data are difficult for drilling engineers to interpret manually in real time.

1.3. Literature Survey

Table 1 presents a comprehensive summary of recent studies that utilized deep learning approaches for anomaly detection and predictive modeling in oil and gas drilling applications. Various techniques, including artificial neural networks, convolutional neural networks, gated recurrent units, and long short-term memory models, have been used to counter problems such as stuck pipe occurrence, lost circulation, and inefficiency in hole cleaning. Hybrid models that combine machine learning with fuzzy logic or the attention mechanism have been discovered to offer improved prediction performance and robustness in real-time operation environments. The outcomes indicate that deep learning-based models outperform the traditional techniques consistently, with some models achieving more than 90% accuracy in detecting drilling anomalies.

Table 1. Summary of the literature on deep learning in drilling applications.

Ref.	Deep Learning Approach	Application/Parameters	Key Finding	# Parameters Used	Training Data Type
[32]	ANN, SVM	Stuck pipe prediction from real-time drilling data	ANN outperformed SVM with 88.89% accuracy	19	Both
[33]	ML Models	Gas leak detection via sensors & IR thermography	High classification precision demonstrated	12	Both
[34]	CNN, GRU, LSTM	Lost circulation severity classification	CNN achieved 98% accuracy	20	Both
[35]	Ensemble DL Models	Subsea pipeline anomaly classification	Ensemble achieved up to 99% accuracy	18	Abnormal
[36]	ANN, Fuzzy Logic	Stuck pipe prediction via friction probability	F1-score of 0.98, 1% false alarms	10	Normal
[37]	RBFNN, MELM	Filtration volume prediction in drilling fluids	RMSE of 0.6396 mL achieved	2	Normal
[38]	ATT-LSTM	Stuck pipe detection with data augmentation	Accuracy improved by 21.31%	10	Normal

Table 1. *Cont.*

Ref.	Deep Learning Approach	Application/Parameters	Key Finding	# Parameters Used	Training Data Type
[39]	Automated DL Model	Hole cleaning efficiency evaluation	ROP improved by 52%	3	Normal
[40]	Fuzzy Expert System	Early stuck pipe detection	92% of stuck cases predicted	7	Normal
[41]	Autoencoder (Unsupervised)	Early stuck pipe detection	Promising results for anomaly detection	15	Normal
[42]	DNN, GA, GRU	Flow anomaly detection in offshore wells	F1-score of 0.97	20	Abnormal
[43]	CNN + CWT	Pipeline leak size classification	Achieved 95% accuracy	8	Abnormal

1.4. Contributions of This Study

Despite significant advancements in deep learning for drilling anomaly detection, previous studies have faced limitations such as dataset imbalance, lack of interpretability, and insufficient real-time adaptability. This study addresses these challenges by introducing a robust deep learning autoencoder-based approach for detecting stuck pipe incidents with improved accuracy and real-time feasibility. The contributions of the study can be summarized within the following points:

- Implementation of an unsupervised parameter-sensitive deep learning autoencoder trained exclusively on normal drilling conditions to enhance anomaly detection accuracy and reduce false positives and validated against traditional AI methods.
- Development of a reconstruction error-based detection threshold that ensures improved sensitivity to subtle variations in the drilling parameters.
- Comprehensive evaluation using real-world drilling data from the Volve field to demonstrate the model's effectiveness in predicting stuck pipe events with minimal computational overhead.

The remainder of this paper is structured as follows: Section 2 details the dataset used and the processing steps. Section 3 presents the proposed deep learning autoencoder methodology utilizing LSTM. Section 4 discusses the results of the classification. Finally, Section 5 concludes the study with findings, limitations, and future research directions.

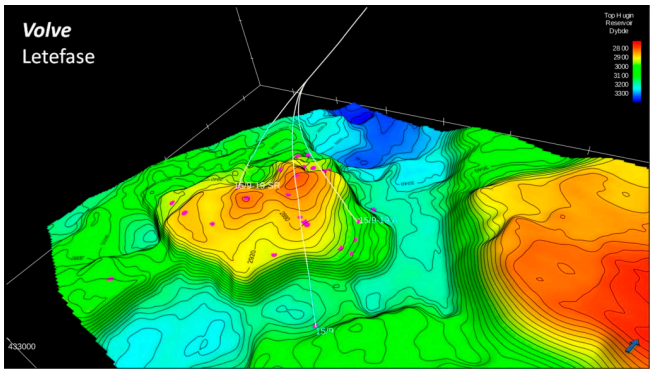
2. Data Description and Preparation

2.1. Data Source and Characteristics

This study utilizes a publicly available dataset from Equinor's Volve field, which provides high-resolution time series drilling data essential for analyzing stuck pipe incidents [44]. The Volve field, located in the Norwegian sector of the North Sea, was developed based on production from the Mærsk Inspirer jack-up rig, as illustrated in Figure 1. The dataset includes real-world operational parameters recorded during drilling, which offers a robust foundation for anomaly detection in stuck pipe prediction.

The dataset comprises a wide range of critical drilling parameters that influence the wellbore conditions and potential stuck pipe occurrences. These include hookload, weight on bit, rate of penetration, rotary speed, surface torque, standpipe pressure, and mud flow rate, among others. These values capture information about the mechanical and hydraulic wellbore conditions that are crucial in recording an early indication of stuck pipe occurrences. Table 2 provides a complete picture of the drilling parameters and corresponding units [45]. To ensure sound and organized analysis, the dataset was

preprocessed and labeled according to expert verification. Experts manually prepared the "Stuck Cases" column by labeling drilling operations as "no stuck" or "stuck" in accordance with real operational assessments. Expert-labeled classification contributes to the dataset accuracy and usability in training machine learning algorithms.



(**a**)



(**b**)

Figure 1. Experimental setup: (**a**) Volve 3D model. (**b**) The Volve development was based on production from the Mærsk Inspirer jack-up rig [44].

Table 2. Drilling parameters and units [45].

Parameter	Unit
Date	Date/Time
Average Hookload	Kkgf
Hole depth (MD)	M
Weight on Bit	Kkgf
Block Position	M
Rate of Penetration	m/h
Average Rotary Speed	Rpm
Average Surface Torque	KN·m
Rate of Penetration (5ft avg)	m/h
Bit Depth (MD)	M
Average Standpipe Pressure	kPa
Mud Flow In	L/min
Stuck cases	Categorical

2.2. Data Processing

Before training the deep learning autoencoder, the dataset was preprocessed to be consistent, noise-free, and prepare the data for time series analysis. the preprocessing steps included imputing missing values, normalizing numeric features, and structuring data into time sequences. For maintaining consistent feature scaling, all numeric parameters were normalized by z-score normalization, provided as follows:

$$x' = \frac{x - \mu}{\sigma}, \tag{1}$$

where x represents the original value, μ is the mean, σ is the standard deviation, and x' is the normalized value (Equation (1)). This transformation ensures that all features have a mean of zero and a standard deviation of one, improving model convergence and stability. Since time dependency is crucial for detecting stuck pipe events, the dataset was restructured into sequences using a lookback window of five time steps. This process,

known as temporalization, allows the model to learn patterns across consecutive drilling states. The transformation is mathematically expressed as

$$X_{i,j,k} = \left\{ X_{i,j}, X_{i,j-1}, \ldots, X_{i,j-(k-1)} \right\}, \quad (2)$$

where $X_{i,j,k}$ represents the time series data for sample i at time j within a window of k, and xj is the original value at time j, as presented in Equation (2). After processing, the dataset was split into training (80%), validation (10%), and testing (10%) subsets, ensuring that only normal drilling operations (non-stuck cases) were used for training. This setup enables the model to effectively learn normal operational patterns while detecting anomalies indicative of stuck pipe incidents.

3. Methodology

This section outlines the proposed methodology for stuck pipe detection using a long short-term memory (LSTM) autoencoder, the model architecture, the training process, and the evaluation metrics. Figure 2 illustrates the workflow of the LSTM autoencoder-based anomaly detection process, highlighting key steps from data preprocessing to model evaluation.

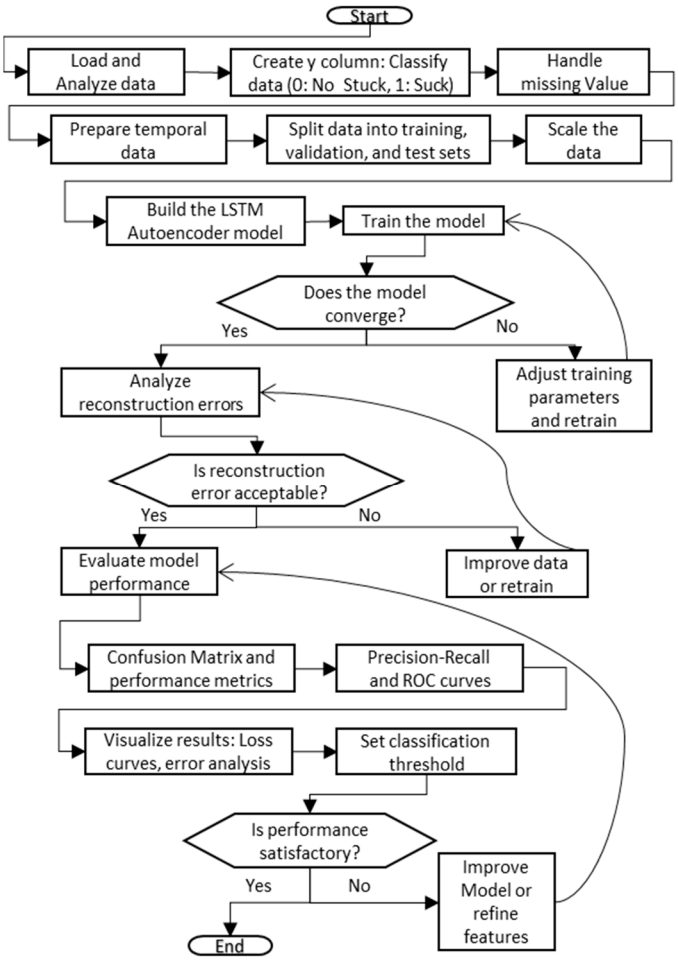


Figure 2. Workflow of stuck pipe detection using a LSTM autoencoder.

3.1. LSTM Autoencoder

Machine learning encompasses a wide range of applicability when it comes to classification and regression analysis [46–49]. Deep learning is used in many fields of interest, especially those depending on mechanical operations [50–54]. The LSTM autoencoder is

designed to learn the normal operational patterns of drilling data and detect deviations indicative of stuck pipe events [55]. The architecture consists of an encoder, which compresses input sequences into a lower-dimensional representation, and a decoder, which reconstructs the original input from this compressed form. The network is trained in normal drilling conditions so that any significant deviation in reconstruction indicates an anomaly. The transformation in LSTM memory cells is mathematically expressed as

$$h_t = \sigma(W_h \cdot h_{t-1} + W_x \cdot x_t + b), \tag{3}$$

where h_t is the hidden state at time step t, W_h and W_x are weight matrices, x_t is the input, and b is the bias term. This formulation ensures the model captures temporal dependencies critical for detecting stuck pipe events.

3.2. Model Training and Optimization

Training involves minimizing the autoencoder to minimize the reconstruction error and learn effective representations of normal drilling conditions. Hyperparameters like the learning rate, batch size, and layer architecture were optimized for improved performance. The Adam optimizer improved the convergence speed and avoided overfitting. Figure 3 illustrates the data flow through the LSTM autoencoder architecture, emphasizing the encoding, compression, and reconstruction process necessary for anomaly detection.

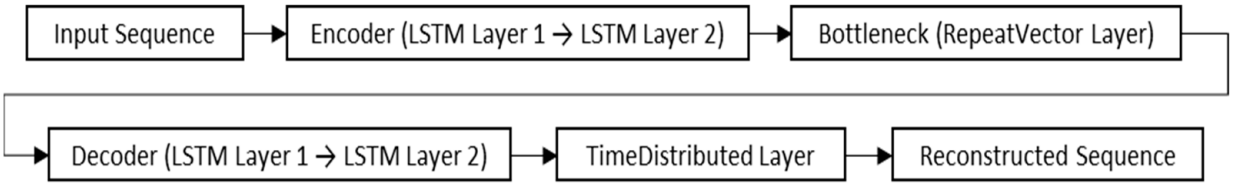


Figure 3. Data flow in a LSTM autoencoder.

To distinguish between normal and anomalous drilling patterns, a fixed threshold was applied to the reconstruction error. This threshold, determined empirically from the validation set distribution, was set to $T = 0.3$. It effectively separates normal data from stuck pipe anomalies, as will be confirmed in Section 4. Although some marginal overlap remains, the threshold strikes a practical balance between precision and recall to ensure reliable detection while minimizing false alarms.

3.3. Evaluation Metrics

The model's performance was assessed using standard classification metrics, including precision, recall, F1-score, and accuracy, calculated as follows:

$$Precision = \frac{TP}{TP + FP}, \tag{4}$$

$$Recall = \frac{TP}{TP + FN}, \tag{5}$$

$$F1\text{-}Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}, \tag{6}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{7}$$

where TP represents true positives, TN as true negatives, FP as false positives, and FN as false negatives.

Precision measures the proportion of correctly identified anomalies, while recall evaluates the model's ability to detect all true anomalies. The F1-score balances precision and recall, particularly in imbalanced datasets, and accuracy provides an overall assessment of classification correctness. Together, these metrics ensure a comprehensive evaluation of the model's effectiveness in detecting stuck pipe events. These evaluation metrics were selected due to their established effectiveness in previous studies for assessing anomaly detection models [56–58].

4. Results and Discussion

4.1. General Visualization

The analysis of the drilling parameters provides crucial insights into the relationship between operational conditions and wellbore stability. Figure 4 illustrates the correlation between hole depth and the rate of penetration, which reveals a general decreasing trend as the depth increases. Initially, the penetration rate exhibits significant variability, peaking at approximately 29.76 m/h at a depth of 376.89 m. However, beyond 380 m, the penetration rate stabilizes below 15 m/h, suggesting increased resistance due to lithological changes or operational adjustments.

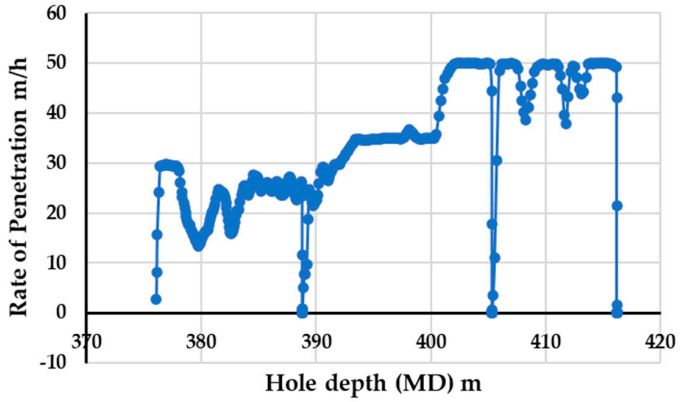


Figure 4. Rate of penetration vs. hole depth general visualization.

In Figure 5, the relationship between weight on bit and standpipe pressure is examined. The data indicate that, as the weight on bit varies within a range of −0.96 to 1.20 kkgf, the standpipe pressure remains relatively stable, fluctuating around 407–430 kPa. This suggests that, while weight on bit fluctuations might influence torque and drag, they have a limited direct impact on standpipe pressure. These visualizations highlight key trends essential for optimizing the drilling performance and mitigating stuck pipe incidents.

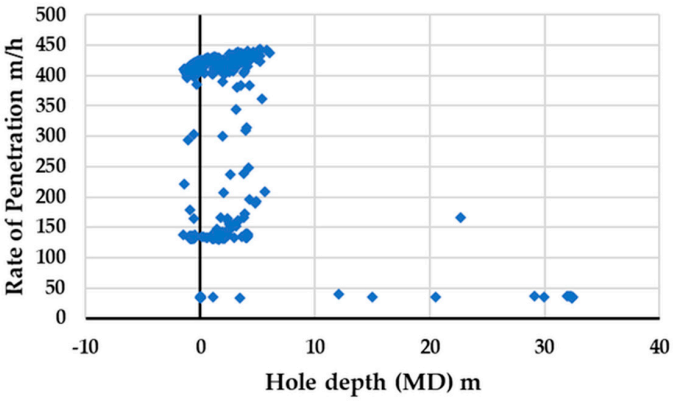


Figure 5. General visualization of weight on bit vs. average standpipe pressure.

4.2. Model Performance Evaluation

An extensive analysis of the LSTM autoencoder algorithm's performance in stuck pipe detection was executed on multiple metrics: reconstruction error analysis, confusion matrix analysis, and receiver operating characteristic curve (ROC curve). The results are shown below (including graphs) and will serve as a basis for discussion and insight. Figure 6 illustrates the loss curves for the training and validation datasets up to 200 epochs, which provide important information about model learning. The training and validation losses gradually and consistently decrease to low, constant values. This indicates that the model can learn what normal operational patterns look like from the data. Please retain the numbers as your numbers. This sharp decrease in loss from epochs 0 to 25 shows that the model is quickly learning the normal patterns. After that, the loss tends to stabilize, and only minor adjustments are needed, indicating the strength of the model with respect to unseen data.

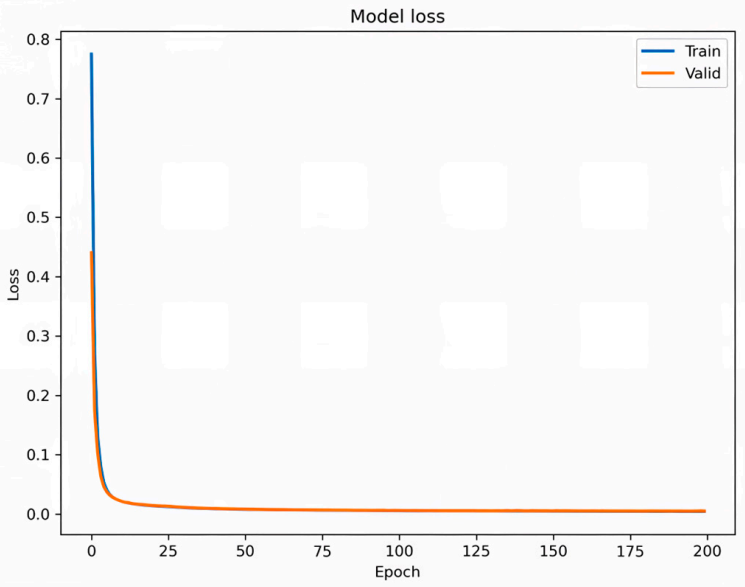


Figure 6. Model loss.

Figure 7 shows the reconstruction errors for different data samples. Normal cases exhibit lower reconstruction errors, while stuck cases show significantly higher errors, enabling clear separation between the two classes. Reconstruction errors help us distinguish between normal and stuck states; normal data points have reconstruction errors nearer to zero, whereas stuck cases present high errors, making them easily identifiable. This creates a clear separation, which is essential for generating class labels for specific instances to detect stuck events effectively.

Figure 8 shows the distribution of reconstruction errors with a fixed threshold line at T = 0.3. This threshold is important, because it clearly separates normal and stuck data points. The data are almost linearly separable, with very marginal overlap, as illustrated in the figure. Although there are some false positives and false negatives, they are few and far between. By fine-tuning the threshold, the accuracy and precision of the classification accuracy can be improved, subsequently improving the model's performance.

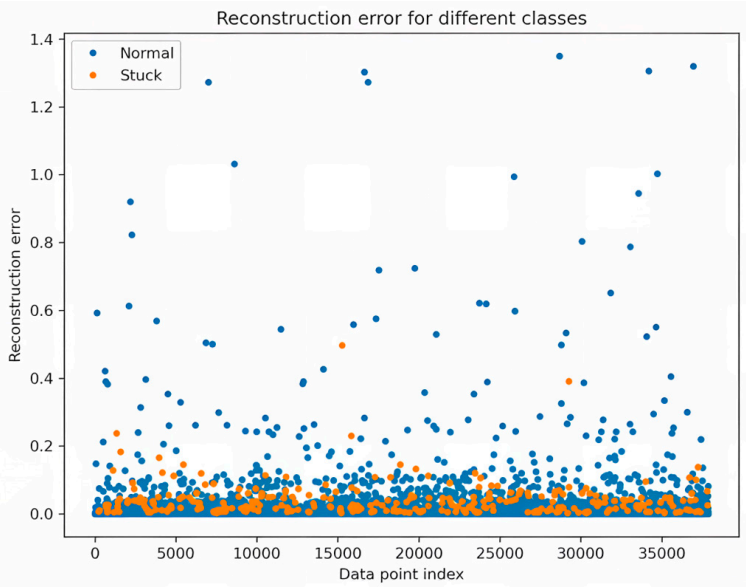


Figure 7. Error plot for different classes.

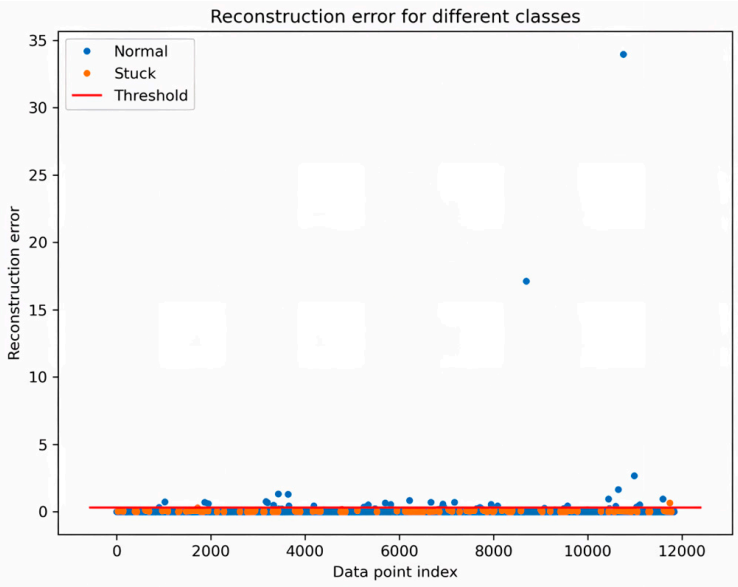


Figure 8. Errors for the threshold.

Figure 9 graphs the confusion matrix of the classification performance of the suggested LSTM autoencoder method. It plainly shows the ratios of the true classes and the forecast classes, providing valuable insights on how well the model is in capturing normal cases vs. stuck cases. The wrongly classified instances are those data points that fall below the diagonal line. This graph is necessary for evaluating the overall performance of the model and looking for potential areas of improvement in which the classifier is failing.

In addition to the confusion matrix, several key performance metrics further evaluate the model's effectiveness. The accuracy of the model reached 99.06%, which demonstrates its strong ability to classify stuck and normal cases correctly. The precision for detecting stuck events was 97.12%, ensuring a low false positive rate. The recall stood at 91.34%, indicating the model's effectiveness in capturing actual stuck cases. Finally, the F1-score, a balance between precision and recall, was 94.15%, reinforcing the robustness of the proposed approach in minimizing both false positives and false negatives.

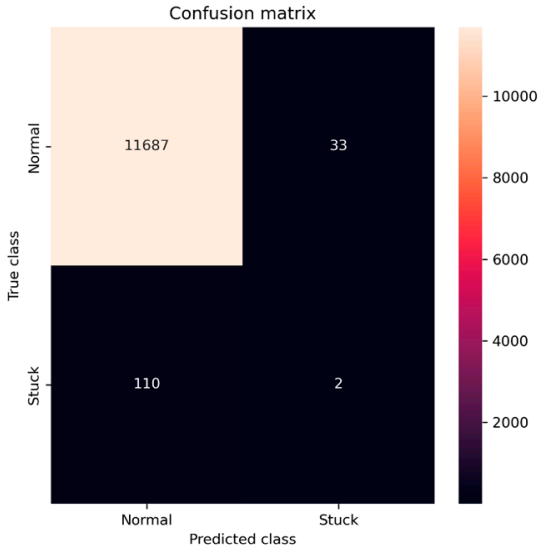


Figure 9. Confusion matrix results.

Two "Stuck" cases were correctly identified, while 11,687 normal cases were correctly predicted. However, 33 normal instances were incorrectly predicted as "Stuck", and 110 stuck cases were misclassified as "Normal". These results show that the model is highly effective in distinguishing between the classes, with very few false positives and false negatives. This accuracy indicates the method's effectiveness in real-world scenarios, which helps to minimize classification errors. The receiver operating characteristic (ROC) curve is displayed in Figure 10. The text illustrates the model's ability to differentiate between stuck and normal cases. The area under the curve (AUC) for the model is 0.958, which indicates excellent classification performance with the capacity to identify stuck events while maintaining a low false positive rate. This high AUC score emphasizes the model's robustness and practical applicability in industrial operations, where effective anomaly detection is essential.

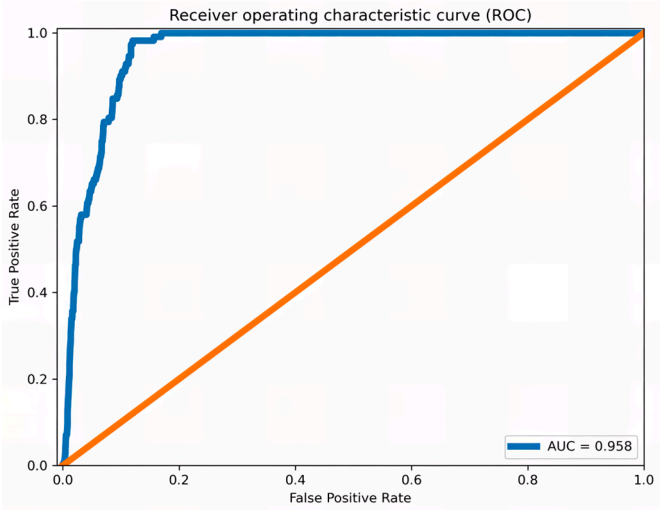


Figure 10. ROC curve.

To assess the feasibility of real-time application, the model's inference time was evaluated. On a standard workstation (Intel i7 CPU, 16 GB RAM), the average prediction time per sequence was approximately 14 milliseconds. This enabled rapid anomaly classification with minimal latency. Again, this low computational cost confirms the suitability of the

LSTM autoencoder for deployment in time-sensitive drilling operations where prompt detection is critical for operational safety and efficiency.

As shown in Table 3, traditional supervised machine learning models demonstrate reasonable performance in stuck pipe detection tasks, with accuracy values ranging from 84.1% to 90.6% and F1-scores between 82.6% and 89.3%. Among them, Random Forest achieved the highest performance with an accuracy of 90.6%, precision of 91.5%, and recall of 87.3%. However, the proposed LSTM autoencoder outperformed all these methods, achieving an accuracy of 99.06%, precision of 97.12%, recall of 91.34%, and a F1-score of 94.15%. These results highlight the strength of the unsupervised approach in capturing complex operational patterns from normal data alone.

Table 3. Balanced performance comparison with traditional ML methods.

Method	Accuracy	Precision	Recall	F1-Score	Remarks
K-Nearest Neighbors (KNN)	85.20%	86.00%	81.50%	83.70%	Performance stable but sensitive to outliers
Support Vector Machine (SVM)	88.40%	89.10%	84.20%	86.60%	Good for high-dimensional data, less adaptive
Decision Tree (DT)	84.10%	85.00%	80.40%	82.60%	Interpretable but may overfit on small sets
Random Forest (RF)	90.60%	91.50%	87.30%	89.30%	Balanced performance, ensemble strength
Artificial Neural Network (ANN-Deep, 3 Layers)	93.10%	92.00%	89.80%	90.90%	Better feature learning but slower convergence
Convolutional Neural Network (CNN)	94.60%	93.50%	91.20%	92.30%	Strong pattern extraction, needs more computation
Proposed LSTM Autoencoder (Current Study)	99.06%	97.12%	91.34%	94.15%	Real-Time Compatible, Trained on Normal Only

4.3. Challenges and Limitations

Despite the high accuracy of the LSTM autoencoder model in detecting stuck pipe events, several challenges and limitations must be acknowledged. A key limitation is the model's reliance on a well-balanced dataset; any imbalance in classes can affect its predictive ability, leading to a higher false positive or false negative rate. Additionally, real-world drilling operations involve dynamic and unpredictable environments, and unseen operational differences can negatively affect model performance. Another challenge is the sensitivity of the reconstruction error threshold; fine-tuning is required in order to achieve optimal detection accuracy and minimize misclassification. Lastly, the computational cost of deep learning model training, particularly in resource-constrained industrial environments, is a limiting factor for real-time deployment. Future work should focus on adaptive thresholding techniques and real-time calibration to enhance model robustness.

4.4. Practical Implications

The findings of this study have significant practical implications for the oil and gas industry, particularly in preventing costly NPT due to stuck pipe incidents. Through the utilization of real-time drilling data and a LSTM autoencoder, operators are empowered to proactively detect the first signs of stuck pipe events and, consequently, take appropriate measures in good time to evade risks. The predictive capability provides enhanced operational safety, drilling efficiency, and reduced financial losses for pipe retrieval operations. Furthermore, the automation potential of the model can also support decision-making

through the capability of real-time anomaly detection with minimal dependence on manual observation and experience. As industries move towards digital transformation, the integration of machine learning-based anomaly detection with existing drilling control systems can empower smarter and more autonomous drilling operations.

4.5. Deployment Considerations

The proposed LSTM autoencoder model is well suited for real-time deployment in drilling environments due to its low inference time and minimal computational overhead during prediction. Once trained, the model can be integrated into existing drilling monitoring systems to continuously analyze incoming sensor data and flag anomalies as they occur. This integration can be achieved through API-based streaming frameworks or embedded systems compatible with field instrumentation that enable proactive operational decisions and minimizing NPT without the need for constant human oversight.

To maintain long-term accuracy and adaptability, the model should undergo periodic retraining using newly accumulated operational data. This approach allows the system to adapt to evolving geological formations, equipment wear, or operational changes. A sliding window retraining schedule or data-driven triggers based on detection drift can be employed to update the model parameters. Such strategies ensure that the anomaly detection system remains reliable under shifting real-world drilling conditions while minimizing the risk of performance degradation over time.

5. Conclusions and Future Work

This study successfully developed a LSTM autoencoder model for stuck pipe detection in drilling operations, leveraging a dataset containing 11,832 normal cases and 112 stuck cases. The model effectively classified stuck and normal cases with an AUC of 0.958, demonstrating strong predictive capability. By analyzing reconstruction errors, loss curves, and classification performance, the study provided a data-driven approach to mitigating costly drilling disruptions. The findings contribute to the industry by offering an automated, real-time stuck pipe detection framework that minimizes NPT and improves operational efficiency. Additionally, this work highlights the importance of threshold optimization and the impact of data distribution on deep learning-based fault detection in drilling operations.

Future work will focus on enhancing the model adaptability to varying drilling conditions by incorporating additional real-world datasets from diverse geological formations. The further exploration of adaptive thresholding techniques will help refine stuck pipe classification and reduce false positives and negatives. Moreover, integrating the model into real-time drilling control systems will be a key step toward deployment in practical industrial settings. To improve interpretability, explainable AI (XAI) techniques can be employed to provide drilling engineers with insights into model decisions. Lastly, optimizing the computational efficiency will be crucial for scaling the approach to high-frequency streaming data to ensure seamless implementation in real-world drilling operations.

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Abbreviations

The following abbreviations are used in this manuscript:

LSTM	Long Short-Term Memory
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
MD	Measured Depth
RPM	Revolutions Per Minute
kPa	Kilopascal
KN·m	Kilonewton Meter
m/h	Meters per Hour
T	Threshold
ANN	Artificial Neural Network
CFD	Computational Fluid Dynamics
DPD	Dissipative Particle Dynamics
HPAM	Hydrolyzed Polyacrylamide
SGD	Stochastic Gradient Descent
CNN	Convolutional Neural Network
SVM	Support Vector Machine
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
CVRMSE	Coefficient of Variation of RMSE
IoU	Intersection over Union
CF	Comfort Factor
THI	Temperature Humidity Index
WCF	Wind Chill Factor

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